Why Y is 10^11 Pa: the interatomic spring

One bond energy over one atomic volume sets the stiffness of every ordinary solid.

A chain of bonds under load

$$F = U''(r_0)\, s, \qquad \sigma \approx \frac{F}{r_0^2}, \qquad \varepsilon = \frac{s}{r_0}$$

$$Y = \frac{\sigma}{\varepsilon} = \frac{U''(r_0)}{r_0}$$

Model neighbours by a pair potential U(r) with its minimum at r0. Stretch one bond by s: the restoring force is the curvature times s. One chain of bonds occupies an area of order r0^2 and the strain is s/r0, so the macroscopic modulus is the bond curvature divided by the bond length.

WHY / This is the microscopic content of Hooke law: the linear term in the potential becomes the linear relation between stress and strain. Curvature, not depth, is the spring constant.

WATCH / The estimate assumes a simple pair potential and a cubic lattice. Directional covalent bonds and long-range ionic forces make the real modulus anisotropic, so this is an order-of-magnitude argument, not a prediction.

Why the answer is 10^11 Pa

$$U''(r_0) \sim \frac{D}{r_0^2} \;\Rightarrow\; Y \sim \frac{D}{r_0^3}$$

$$D = 3\ \text{eV} = 4.8 \times 10^{-19}\ \text{J},\ r_0 = 2\ \text{Å}:\ Y \sim 6 \times 10^{10}\ \text{Pa}$$

A well of depth D and width r0 has curvature of order D/r0^2, so the modulus is D/r0^3: an energy per volume. Bond energies are a few electronvolts and bond lengths a couple of angstroms, and there is no other combination available, which is why every ordinary solid sits between 10^10 and 10^11 Pa.

WHY / The same estimate gives the interatomic spring constant ka = Y r0, about 40 N/m for steel: a laboratory-scale spring constant hidden inside a bond.

WATCH / Rubber at 10^6 Pa is not a failure of the arithmetic. Its restoring force is entropy of coiled chains, not the curvature of a well, and the estimate simply does not apply.

The interatomic force constant

$$k_a = U''(r_0) = Y r_0$$

$$\omega_{\text{vib}} = \sqrt{k_a/m} \sim 10^{13}\ \text{s}^{-1}$$

The curvature that gives Y also gives one bond spring constant: about 40 N/m for steel with r0 near 2 angstrom. Dividing by an atomic mass gives a vibration frequency near 10^13 per second, the Debye scale.

WHY / This is the numerical bridge between a modulus and a microscopic spring. It also explains why sqrt(Y/rho), the elastic wave speed, is of the order of the speed of sound in the solid.

WATCH / Do not insert this bond stiffness into a macroscopic formula without converting: a rod of length L has of order L/r0 bonds in series, which is why the macroscopic stiffness carries A/L.

Compressibility from the same picture

$$B = -V\frac{dp}{dV} \sim \frac{D}{r_0^3}\ \text{for a close-packed solid}$$

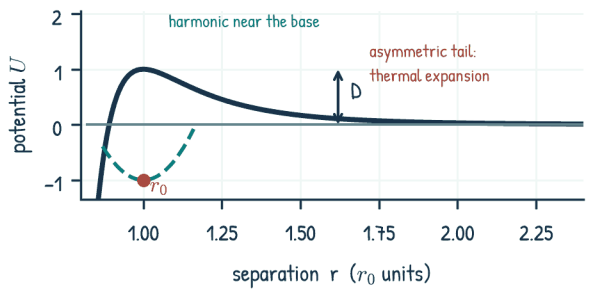
$$B_{\text{steel}} \approx 1.6 \times 10^{11}\ \text{Pa}\quad (Y = 200\ \text{GPa},\ \nu = 0.29)$$

Squeezing a solid changes every bond length at once, so the bulk modulus is again an energy per volume. For steel, B is close to Y but slightly smaller; for rubber, B is three orders of magnitude larger than Y, which is the defining feature of a nearly incompressible material.

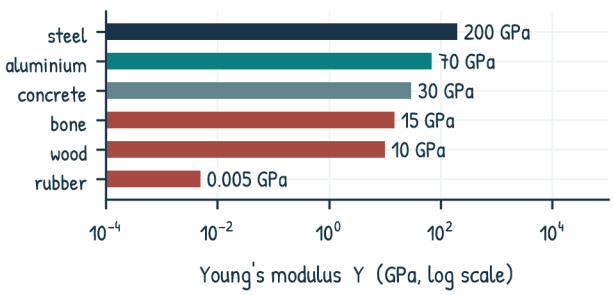
WHY / Order-of-magnitude estimates for B and Y agree because both come from the same bond; the ratio between them is set by Poisson ratio, not by chemistry.

WATCH / Do not apply the solid estimate to a liquid. Water has B = 2.2 GPa because its hydrogen-bond network is already partly broken; its shear modulus is zero because molecules slide.

Well curvature sets Y; well asymmetry sets thermal expansion



Order-of-magnitude smell test: solids 10–200 GPa, rubber 10^-3 GPa



REMEMBER / Moduli are bond energies per atomic volume; no other scale is available.

Thermal expansion as the asymmetry of a well

A symmetric well cannot expand; the cubic term is the whole phenomenon.

The classical average

$$U = \tfrac{1}{2}ks^2 + \tfrac{1}{6}\gamma s^3, \qquad \gamma = U'''(r_0) < 0$$

$$\langle s \rangle = -\frac{\gamma k_B T}{2k^2}$$

Write the well as a parabola plus a cubic term, with gamma negative for a well that is softer outward. Average s over the Boltzmann distribution. The correction to the mean comes from the even part of the cubic-weighted integrand, and it is linear in T.

WHY / Harmonic vibrations alone give zero expansion: the mean position stays at r0 at every amplitude. Only the asymmetry of the well converts vibration into a drift, so expansion is a direct measurement of the cubic term.

WATCH / The sign of gamma matters. A well that is steeper outward would contract on heating, and a few real materials do over restricted ranges. The estimate below assumes the usual softer-outward shape.

Collapse to alpha about k_B / 2D

$$\alpha = \frac{1}{r_0}\frac{d\langle s \rangle}{dT} = -\frac{\gamma k_B}{2r_0 k^2}$$

$$|\gamma| \sim \frac{k}{r_0}, \quad k \sim \frac{D}{r_0^2} \;\Rightarrow\; \alpha \sim \frac{k_B}{2D}$$

Estimate the cubic coefficient as k/r0 and the curvature as D/r0^2. Everything cancels except Boltzmann constant over bond depth, so alpha is about 1.4e-5 per kelvin for a 3 eV bond, which is the measured order for metals.

WHY / The same relation explains the correlation between high modulus and low expansion: a deep, stiff well expands less. Diamond is stiff and expands very little; rubber is soft and expands a lot.

WATCH / The estimate is a lower-order classical result. Quantum zero-point motion and the full phonon spectrum change the coefficient by factors of order unity, not orders of magnitude.

Constrained expansion and stress

$$\sigma_{\text{th}} = Y\alpha\Delta T, \qquad \Delta L_{\text{free}} = \alpha L \Delta T$$

$$\alpha B \approx \frac{k_B}{\text{atomic volume}}$$

A solid that cannot expand converts the prevented strain into stress, and Y alpha is fixed by the same bond parameters: about 2.4 MPa per kelvin for steel, so a 100 K constraint is already dangerous.

WHY / The relation alpha B approximately constant per atomic volume ties thermal and elastic response to one anharmonicity: two measurements of a single microscopic quantity.

WATCH / Expansion is not a force and does not accumulate over cycles. A rail that breathes freely for a year develops no stress at all; only the constrained part of the swing produces stress.

Where the model stops

$$\text{Invar}: \ \alpha \approx 10^{-6}\ \text{K}^{-1} \ \text{by magnetic compensation}$$

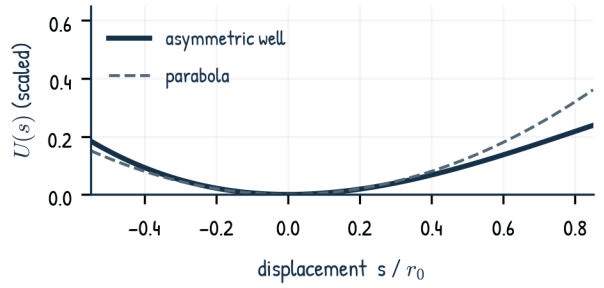
$$\text{anisotropic crystals}: \ \alpha \text{ is a tensor}$$

Real solids can have alpha near zero (Invar) because a magnetic contribution cancels the phonon one, and anisotropic crystals expand differently along different axes. The scalar, monotone alpha of this model is a first approximation.

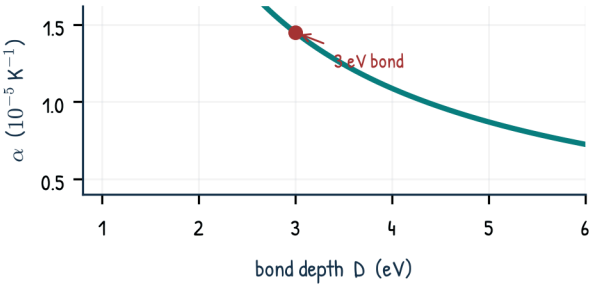
WHY / The existence of Invar is a warning: a material property that seems fundamental can be engineered by adding a second mechanism of the opposite sign.

WATCH / No simple pair-potential argument predicts Invar. Quoting alpha = k_B/2D there would be wrong by an order of magnitude, so state the assumption when you use the estimate.

A cubic term makes the well softer outward and the mean separation drift



Deeper well, smaller expansion: alpha about k_B / 2D



REMEMBER / Expansion is the drift of the mean separation, not a change in the vibration amplitude alone.

The falling weight with the rod own mass

Keep the rod mass in the energy ledger and the answer more than doubles.

Energy with a distributed mass

$$\int_0^L \tfrac{1}{2} Y A \left(\tfrac{du}{dx}\right)^2 dx = \tfrac{1}{2} k \delta^2 \quad (u = \delta x / L)$$

$$\tfrac{1}{2} k \delta^2 = Mg(h+\delta) + \tfrac{1}{2} mg\delta$$

Assume the rod deforms in its static shape u(x) = delta x/L, which is also the shape that minimises the elastic energy at a given end displacement. The rod centre of mass descends delta/2, so its own weight contributes half of mg delta. Every velocity in this self-similar mode is zero at maximum extension.

WHY / The static shape is a variational choice, not an extra assumption about dynamics: it gives an upper bound on the true kinetic energy and becomes exact for slow loading.

WATCH / The rod weight is not simply added to M. It enters at half the end displacement because its centre of mass moves half as far. Adding m to M overestimates the effect.

Solve the quadratic

$$k\delta^2 - 2(M + \tfrac{m}{2})g\delta - 2Mgh = 0$$

$$\delta = \frac{(M + m/2)g + \sqrt{[(M+m/2)g]^2 + 2kMgh}}{k}$$

The positive root is the physical one; the negative root corresponds to a state above the release point. With M = 10 kg, m = 40 kg, k = 1e5 N/m and h = 1 mm, the answer is 6.2 mm against 2.7 mm when the rod mass is ignored.

WHY / A heavy rod more than doubles the extension in this example, so the massless-rod formula is not a small correction when the rod outweighs the falling mass.

WATCH / The quadratic assumes the rod remains elastic throughout. If the peak stress exceeds yield the stored energy is no longer half k delta^2 for the same peak force, and the elastic equation underestimates the extension.

The wave-transit caveat

$$c = \sqrt{Y/\rho}, \qquad t_{\text{transit}} \sim \tfrac{L}{c}$$

$$2\text{ m steel rod}: \; t_{\text{transit}} \approx 0.4\text{ ms}$$

The static-shape assumption requires the loading time to be long compared with the time a stress wave takes to cross the rod. If a mass strikes the end faster than about 0.4 ms, the far end does not know the impact has happened and the rod does not deform in its static shape.

WHY / A stress wave of amplitude rho c v arrives at the support before the rod has agreed on a shape, and the energy is then partitioned by wave mechanics, not by the static mode.

WATCH / This is a validity condition, not a correction factor. The quasi-static formula can be wrong by a large factor under impact, and no amount of algebra inside the formula will reveal it.

What the oscillating solution adds

$$\omega = \sqrt{k/(M + m/3)}, \qquad \delta(t) = \delta_{\text{eq}}(1 - \cos \omega t)$$

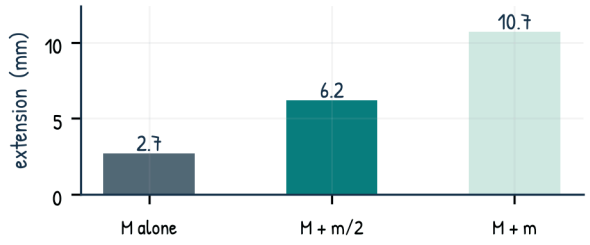
$$\text{effective mass } m/3 \text{ for the static mode}$$

If the rod mass is treated as a lumped correction to the equation of motion rather than through energy, the effective mass is m/3, not m/2. The two results differ because they answer different questions: energy at the turning point versus frequency of oscillation.

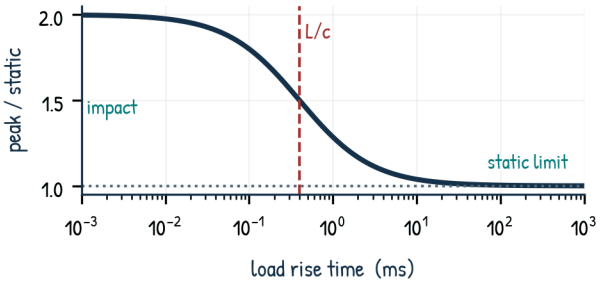
WHY / The discrepancy is a useful reminder that lumping a distributed system is a modelling choice. Quote which quantity the effective mass is meant to reproduce.

WATCH / Do not mix the two factors in one solution. Pick the energy method or the dynamic method and stay inside its assumptions.

Rod weight at half displacement: 6.2 mm, against 10.7 mm if all of m is added



The static-shape energy method needs loading long compared with L/c



REMEMBER / The rod enters at half the end displacement, and the whole method needs slow loading.

The constant-stress rod and the limits of height

Uniform stress needs an exponential flare; a prismatic column has a maximum height.

Derive the profile

$$\sigma_0 A(x) = W + \int_0^x \rho g A(x')\,dx'$$

$$A' = \tfrac{\rho g}{\sigma_0} A \;\Rightarrow\; A = A_0 e^{x/\lambda}, \quad \lambda = \tfrac{\sigma_0}{\rho g}$$

At depth x the section carries W plus the weight of rod above it. Differentiating the balance shows the area grows in proportion to itself, so the profile is exponential with decay length lambda = sigma0/(rho g); the area under the load is W/sigma0.

WHY / A prismatic column of any width crushes under its own weight once its height exceeds lambda, because widening it adds strength and weight in the same proportion. The flare removes that particular ceiling by making stress constant.

WATCH / This is a strength limit, not a stiffness limit. A real tall structure fails by buckling or by the hydraulic limit of a tree long before it reaches lambda.

Numbers and the real ceilings

$$\lambda_{\text{steel}} = \frac{2.5\times10^{8}}{7800\times 9.8} \approx 3.3\ \text{km}$$

$$\lambda_{\text{wood}} \approx 7\ \text{km}, \quad \lambda_{\text{bone}} \approx 10\ \text{km}$$

For structural steel the crushing length is 3.3 km; for wood and bone it is larger because the strength-to-weight ratio is better. No building, tree or bone comes close to its crushing length, so something else is setting the height.

WHY / The crushing length is the answer to a question nobody asked. It bounds a mode of failure that real slender structures avoid by failing earlier in another mode.

WATCH / Substituting yield stress for the working stress in lambda gives an unconservative height. Design uses a safety factor, so the usable lambda is smaller.

What actually limits a tree

$$P_{cr} = \frac{\pi^2 YI}{L^2} \quad \text{(buckling)}$$

$$h_{\text{hydraulic}} \sim 100\ \text{m} \quad \text{(cavitation and embolism)}$$

A tree is limited by water transport: suction in the xylem becomes large enough to nucleate bubbles, and the column of water breaks. This happens near 100 m, seventy times closer than the crushing length of wood.

WHY / The pattern is general: the binding constraint is usually not the one the formula was derived for. Finding the real limit is the engineering skill.

WATCH / The buckling load of a tree depends on its taper and root anchorage, both beyond this model. Quoting the Euler load for a real tree is an order-of-magnitude estimate only.

How much flare, and for which material

$$\text{ground to load}: \frac{A(L)}{A_0} = e^{L/\lambda}$$

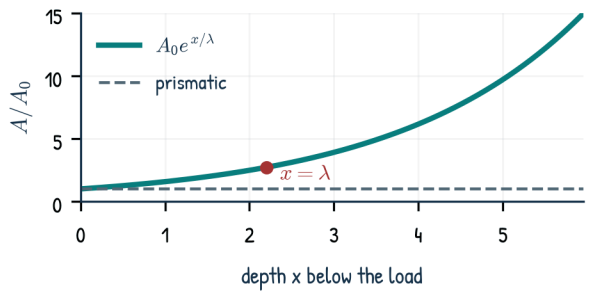
$$L = 30\ \text{m},\ \lambda = 3.3\ \text{km}:\ e^{0.009} \approx 1.01$$

The base is wider than the section under the load by the factor exp(L/lambda), so the flare is visible only when the height approaches lambda. A 30 m steel column flares by about one per cent: a cylinder at building scale, a trumpet only near the crushing length.

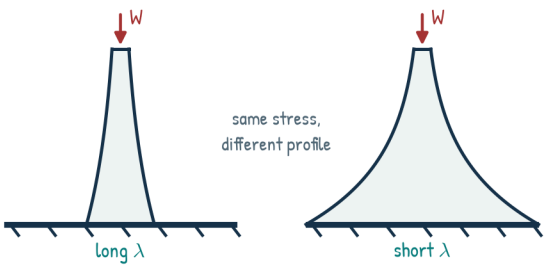
WHY / The exponential is the signature of a balance in which the material carries its own weight: extra area adds strength and load together, and only an exponential keeps their ratio fixed. The exponent is the dimensionless height L/lambda.

WATCH / Do not read exp(L/lambda) as huge at ordinary heights, or attribute a tree taper to this formula: wind, branching and buckling set the shape first.

Constant stress needs an exponential flare; lambda = sigma0/(rho g)



Constant stress: a short decay length needs a violent flare



REMEMBER / lambda = sigma0/(rho g) is a materials length, not the height of a tree.

Bending integrated: slope, deflection and supports

Two integrations, two boundary conditions, one formula per support case.

Curvature to deflection

$$\frac{d^2y}{dx^2} = \frac{M(x)}{YI}, \qquad M = F(L-x)$$

$$y' = \frac{F}{YI}\left(Lx - \frac{x^2}{2}\right), \quad y = \frac{F}{YI}\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right)$$

Take x from the built-in end, y positive downward. Integrate the moment once for the slope and use zero slope at the wall; integrate again for the deflection and use zero deflection at the wall. At x = L the tip deflection is FL^3/3YI.

WHY / The boundary conditions carry the support type. A pinned end has zero deflection but nonzero slope; a built-in end has both zero, which is why the cantilever is the stiffest of the three cases.

WATCH / The small-slope step y double prime approximately equals 1/R is the validity condition. A deflection comparable with L needs the exact curvature, and the cubic formula then underestimates the real deflection.

Uniform load and the comparison

$$M = \frac{w(L-x)^2}{2} \;\Rightarrow\; \delta = \frac{wL^4}{8YI} = \frac{WL^3}{8YI}$$

A uniformly spread load gives a quartic deflection with the same total load W. Comparing FL^3/3YI with WL^3/8YI: the same total load concentrated at the tip deflects 8/3 times as much as when spread along the beam.

WHY / The exponent of L differs because the moment distribution differs. Both results come from the same two integrations; only M(x) changed.

WATCH / A uniform load is not the same as a point load at the centre. For a simply supported beam the centre point load gives FL^3/48YI, and the uniform load gives 5wL^4/384YI, a different coefficient again.

Simply supported and fixed-fixed

$$\text{pinned, centre load}: \; \delta = \frac{FL^3}{48YI}$$

$$\text{fixed}-\text{fixed, centre load}: \; \delta = \frac{FL^3}{192YI}$$

A simply supported beam bends four times as much as a fixed-fixed one under the same centre load. Clamping pushes the points of contraflexure inward, so the beam behaves like a shorter effective span, and its cube supplies most of the factor of four. The cantilever with an end load is the floppiest of the three.

WHY / The support conditions are worth more than a factor of four in deflection while changing nothing about the material or the section. Geometry and constraints dominate.

WATCH / Real joints are neither perfectly pinned nor perfectly fixed. Design codes use effective-length factors between the two ideals, precisely because the boundary condition is uncertain.

Shear, and when bending theory is enough

$$\tau_{\max} = \frac{3}{2}\frac{V}{A} \text{ (rectangle)}, \qquad \delta_{\text{shear}} \sim \frac{VL}{GA}$$

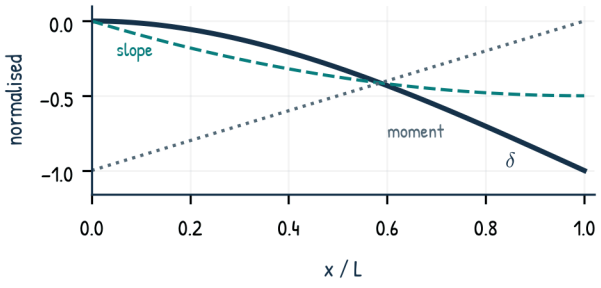
$$\text{slender beam}: \; \frac{\delta_{\text{shear}}}{\delta_{\text{bend}}} \sim \left(\frac{h}{L}\right)^2$$

Bending normal stresses dominate in slender beams, but a short deep beam also deflects by shear. The ratio of shear to bending deflection scales as (h/L)^2, so a beam ten times longer than it is deep is safely a bending problem.

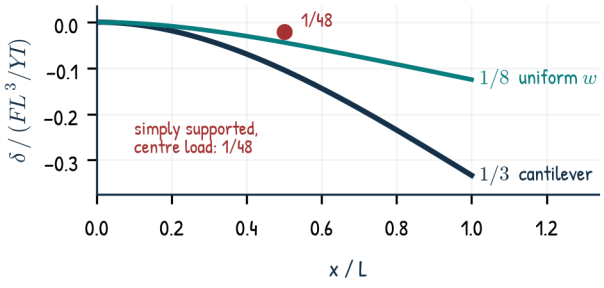
WHY / The same (h/L)^2 ratio decides when the Euler-Bernoulli beam equation is adequate and when Timoshenko corrections are needed. It is a slenderness criterion, not a material one.

WATCH / For a very short stub, bending theory can underpredict the deflection badly. The formula is not wrong; it is being used outside its slenderness range.

Integrate twice: slope from the moment, deflection from the slope



Same load: the cantilever tip sags 16x more than a simply supported centre



REMEMBER / Deflection scales as L^3 and stress as L; they are separate checks.

Euler buckling: an eigenvalue, not a deflection

The amplitude cancels; only the load at which a bent shape is possible survives.

Dimension argument and sine shape

$P \sim \frac{YI}{L^2}$ is the only force built from Y, I and L

$y = \delta \sin\frac{\pi x}{L}, \qquad y'' = -\delta\frac{\pi^2}{L^2}\sin\frac{\pi x}{L}$

A buckling load must be built from the bending stiffness YI and the length; the only combination with the dimensions of force is YI/L^2. For a pinned column the buckled shape is a sine, whose peak curvature is delta pi^2/L^2, and the midspan moment is P delta.

WHY / The amplitude delta cancels from the equation, which is what makes buckling an eigenvalue problem: the column does not choose an amplitude, it chooses the load at which any amplitude is possible.

WATCH / The sine shape assumes pinned ends and a uniform section. Other end conditions change the effective length and therefore the load by the factors 1, 0.25 and 4 shown in the figure.

Euler load and the comparison with yield

$P\delta = YI \cdot \delta\frac{\pi^2}{L^2} \;\Rightarrow\; P_{cr} = \frac{\pi^2 YI}{L^2}$

$P_y = \sigma_y A, \qquad P_{\text{fail}} = \min(P_{cr}, P_y)$

Equating the moment from the axial load to the moment the curvature can carry gives the Euler load. For a 1 cm radius, 2 m steel rod the Euler load is 3.9 kN against a yield load of 79 kN: buckling wins by a factor of twenty.

WHY / Thickening helps as R^4 while shortening helps as 1/L^2, so a column is far more sensitive to its length than to its diameter. That asymmetry is why slender members are dangerous.

WATCH / Euler formula applies only while the material is still elastic. A column that has already yielded buckles at a lower load given by a different, inelastic formula that this chapter does not derive.

Hollow sections and the same mass

$I = \frac{\pi(R^4 - r^4)}{4}, \qquad R^2 - r^2 = \text{const}$

hollow $P_{cr} \approx 3.5 \times$ solid, same mass

At equal mass, moving material away from the axis raises I sharply. A hollow tube of outer radius 1.5 cm and the same area as the 1 cm solid rod has I three and a half times larger and therefore buckles at three and a half times the load.

WHY / This is the same principle as the I-beam and the hollow shaft: the second moment rewards distance from the axis, and the mass only counts area.

WATCH / A thin-walled tube can fail locally by wall buckling before the global Euler load is reached. The wall thickness has a lower bound set by a different instability.

Effective length and real columns

$P_{cr} = \frac{\pi^2 YI}{(\kappa L)^2}, \qquad \kappa = 1,\ 2,\ 0.5,\ 0.7$

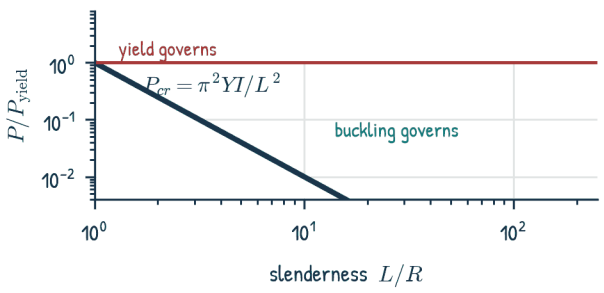
$P \propto \frac{1}{(\kappa L)^2}$: halve κL and carry $4P$

End conditions enter only through the effective length. A fixed-free column behaves as a pinned column of twice the length and carries a quarter of the load; a fixed-fixed column as half the length and carries four times as much.

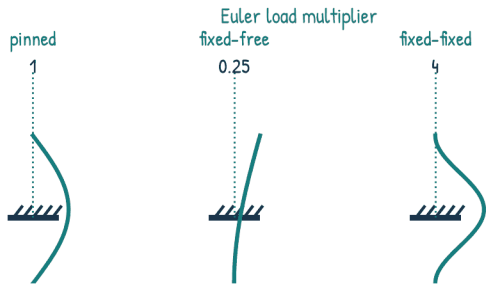
WHY / The effective-length factor is a compact way of saying that rotational restraint is worth as much as shortening the column. Real design codes tabulate kappa for practical joint details.

WATCH / The factors assume ideal end conditions and no sidesway. A frame that can sway has a larger effective length than its members suggest, which is how buildings collapse in a way that looks like a materials failure.

The column fails at the smaller of buckling and yield



End conditions change the Euler load by the factor shown



REMEMBER / A column fails at the smaller of buckling and yield, and slender columns buckle first.

What an I-beam buys, and the thin-walled vessel

Hold the depth fixed before crediting the profile; cut the vessel to get the stress.

The fair I-beam comparison

$$I = \sum \left(\frac{bh^3}{12} + Ad^2 \right)$$

$$\text{rectangle } 20 \times 40: \; I = 1.07 \times 10^5 \text{ mm}^4, \; I/A = 133 \text{ mm}^2$$

For the I-section, flanges contribute mainly through Ad^2 because their own bh^3/12 is negligible at that thickness. A 20x40 rectangle has I/A = 133 mm^2; an 80 mm deep I-section with 680 mm^2 of area reaches about 1000 mm^2, seven times more, but it is also twice as deep.

WHY / Depth alone, entering I as h^2 at fixed area, already accounts for a factor of four. The genuine gain from moving area to the flanges is the remaining factor near two.

WATCH / Quoting five or seven times without holding the depth fixed is the standard exaggeration. Ask what was held constant before believing a factor.

Hoop stress by a diametral cut

$$p \cdot 2rL = 2\sigma tL \;\Rightarrow\; \sigma = \frac{pr}{t}$$

$$p\pi r^2 = \sigma_L \cdot 2\pi rt \;\Rightarrow\; \sigma_L = \frac{pr}{2t}$$

Cut the cylinder along a diameter: the pressure on the projected rectangle 2rL is balanced by two walls of thickness t, giving the hoop stress. Cut across an end cap: the pressure on the circular area is balanced by the ring of wall, giving the longitudinal stress, exactly half as large.

WHY / The hoop stress is twice the longitudinal one, so a thin cylinder always cracks along a generator first. That is why a sausage splits lengthwise and a balloon bursts with a long tear.

WATCH / The thin-wall step assumes the stress is uniform through the thickness. A gun barrel with t comparable to r has a stress gradient through the wall and needs the Lame solution, which this chapter does not develop.

Spherical vessels and the join

$$\text{sphere}: \; \sigma = \frac{pr}{2t} \text{ in every direction}$$

$$\text{cylinder}: \; 2:1 \text{ ratio between hoop and axial}$$

A thin sphere carries the same stress in every direction at half the hoop value of a cylinder of the same r and t, which is why gas storage uses spheres or domed ends. A cylinder-sphere junction has a mismatch in radial growth and therefore a local bending stress.

WHY / Uniform biaxial stress in a sphere means no preferred crack direction and a higher pressure capacity for the same wall. The price is manufacturing difficulty.

WATCH / The junction stress is a discontinuity problem. The membrane formulas describe the fields away from the joint, not the joint itself.

Spider silk and the scale of a strand

$$F = \sigma_f A, \qquad \left.\frac{\sigma_f}{\rho}\right|_{\text{silk}} = 7.7 \times 10^5 \text{ m}^2\text{s}^{-2}$$

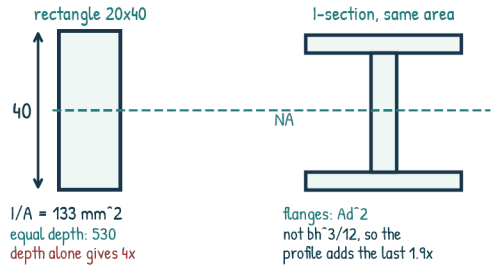
$$\text{steel } 3.2 \times 10^4: \; \text{silk wins by about } 24$$

A 3 micrometre dragline of silk breaks at about 7 mN and can lift less than a gram, while a 70 micrometre strand of the same material lifts hundreds of grams. Specific strength beats steel by a factor of twenty-four, and specific modulus loses to steel by about three.

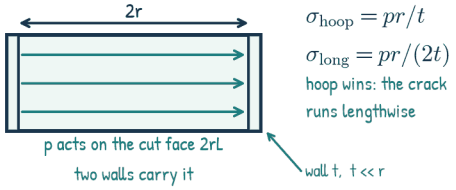
WHY / The strand is an excellent rope and a poor ruler: strength and stiffness are independent axes even at the scale of a protein. Evolution optimised for toughness in tension, not for dimensional stability.

WATCH / Extrapolating one strand to a thick cable ignores statistical defects, which scale with volume. The strength of a long cable is always below the strength of a short fibre.

An I-section puts area where y is large; compare at equal depth and area



Hoop stress by a diametral cut: pr/t, twice the longitudinal stress



REMEMBER / Hoop stress is twice longitudinal, so a pressurised cylinder splits lengthwise.

Two wave speeds and what an earthquake transmits

Each wave speed carries a different modulus under its square root.

Rod, bulk P and S speeds

$$v_{\text{rod}} = \sqrt{\frac{Y}{\rho}}, \qquad v_S = \sqrt{\frac{G}{\rho}}, \qquad v_P = \sqrt{\frac{B + \frac{4}{3}G}{\rho}}$$

A thin rod with free sides carries a longitudinal wave at sqrt(Y/rho). In the bulk the sides cannot contract, so the effective modulus is B + 4G/3 and the wave is faster. A shear wave needs G, so a liquid with G = 0 cannot carry one at all.

WHY / The three speeds are not three materials. They are three different constraints on the same material, and choosing the wrong one is a category error of exactly the kind triage is meant to prevent.

WATCH / Using the rod formula on a seismic arrival underestimates the speed by about 13 % in steel and by more in softer rock. The sides are not free underground.

Reading an earthquake

$$G = \rho v_S^2, \qquad B + \tfrac{4}{3}G = \rho v_P^2$$

$$v_P = 6.0,\ v_S = 3.5\ \text{km/s}: \quad B = 5.3 \times 10^{10}\ \text{Pa}$$

Two measured speeds give two combinations of the moduli, hence G and B and then Poisson ratio. For typical crustal rock the numbers give nu about 0.24, close to steel, because silicate frameworks are elastically similar.

WHY / The inversion from arrival times to moduli is the whole of exploration seismology at its simplest: measure two speeds, infer two moduli, and map the interior.

WATCH / The formulas assume isotropic, homogeneous, linear elasticity. Layering, anisotropy and attenuation all change the inversion, and the numbers quoted are representative averages.

The liquid outer core

$$v_S = 0 \iff G = 0$$

no direct S arrivals beyond 103°

S-waves do not pass through the outer core: beyond a certain angular distance from the source there are no direct shear arrivals, and a shadow zone appears. Since a wave with zero speed requires zero shear modulus, the core must be fluid.

WHY / One absence, two moduli and a wave equation produce a statement about the interior of the Earth that no one has ever sampled directly. This is the strongest inference in the chapter.

WATCH / A fluid core still carries P-waves, so the P shadow is different from the S shadow. Confusing the two shadow zones is the standard error in seismology questions.

Wave speed as a materials tool

$v_{\text{rod}} = \sqrt{Y/\rho}$ is also the sound speed in a bar

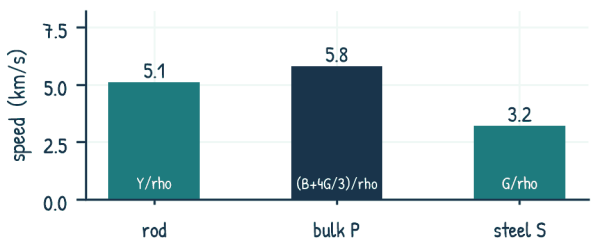
L/c sets the time for a rod to agree on a shape

The same speed that decides an earthquake arrival also decides whether a falling-weight problem may be treated quasi-statically. A 2 m steel rod takes about 0.4 ms to transmit a stress pulse, so impacts faster than that are wave problems, not statics problems.

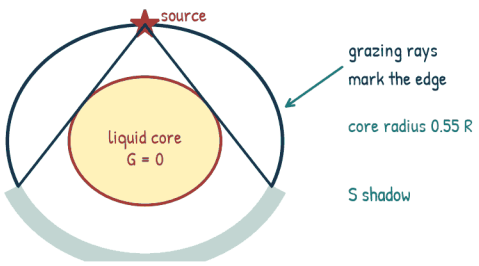
WHY / One number, sqrt(Y/rho), appears in acoustics, seismology and impact mechanics. Recognising it across chapters is more valuable than memorising it three times.

WATCH / The rod speed needs a thin rod with free sides. A bulk sample, a plate and a rod all have different wave speeds for the same material.

Rod, bulk-P and S speeds use different moduli: 5.1, 5.8, 3.2 km/s



The S shadow: rays that would cross the liquid core never arrive



REMEMBER / The absence of S-waves beyond a depth is evidence for a liquid core.

Plane stress, principal axes and Mohr circle

A general in-plane state is three numbers, and two of them can always be made to vanish.

The stress element

$$\sigma_x,\ \sigma_y,\ \tau_{xy} \text{ on a fixed element}$$

$$\text{tension positive; } \tau_{xy} \text{ on the } +\text{x face along } +\text{y}$$

A plane stress state is fixed by three numbers. Normal stresses are positive in tension; the shear sign convention must be fixed once and then used consistently. A general plane has both a normal and a shear component, and rotating the element mixes them.

WHY / Stress is a tensor, not a vector: the traction on a plane depends on the plane. That is why a state described by three numbers transforms under rotation in a way a vector does not.

WATCH / Confusing the component tau_xy with the magnitude of shear on an arbitrary plane is the standard error. The maximum shear is not tau_xy except when the normal stresses are equal.

Principal stresses

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

The principal stresses are the extreme normal stresses; on those planes the shear vanishes. Their sum is invariant under rotation, and the difference sets the maximum shear. The axis angle follows from the transformation.

WHY / Invariance of the trace is the plane-stress version of B depending only on the mean stress: rotation cannot change the part that changes volume.

WATCH / The formula uses the doubled angle because a stress element has 180 degree symmetry, not 360. Forgetting the factor of two puts the principal axes 45 degrees out.

Mohr circle

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}, \qquad C = \frac{\sigma_x + \sigma_y}{2}$$

$$\tau_{\max} = R, \qquad \sigma(\theta) = C + R\cos 2\theta$$

Plot normal stress horizontally and shear vertically. Every plane of the element maps to a point on a circle of radius R centred on the mean stress. The principal stresses are where the circle crosses the axis; the maximum shear is the top of the circle.

WHY / The circle turns the transformation into geometry: a rotation of the physical element by theta moves the point by 2theta around the circle. Nothing is calculated twice.

WATCH / A point inside the circle is not a plane of the element. And the circle is for plane stress only; a three-dimensional state has three circles.

Three dimensions, briefly

$$\sigma_m = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}, \qquad \frac{\Delta V}{V} = \frac{\sigma_m}{B}$$

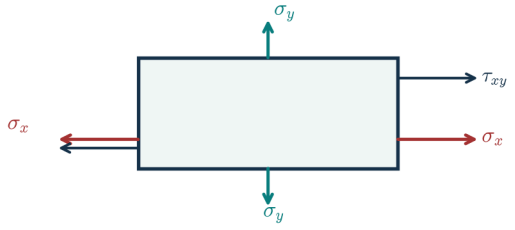
$$\text{deviatoric: shape; mean: volume}$$

In three dimensions a stress state splits into a mean (hydrostatic) part that changes volume and a deviatoric part that changes shape. The bulk modulus responds only to the mean; the shear modulus only to the deviatoric part. This is the exact statement behind the two interrelations.

WHY / The split is why a deep ocean pressure does not shear a fish, and why a pure shear stress does not compress a solid. Each modulus owns one part of the stress.

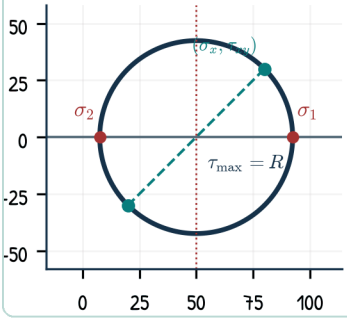
WATCH / A state with all three principal stresses equal is hydrostatic and has zero shear on every plane. This is the only state in which no plane feels shear.

Plane stress: sigma_x, sigma_y and tau_xy, with the sign convention fixed once



three numbers fix the state

Mohr circle to scale: principal stresses are where the circle meets the axis



given
$\sigma_x = 80$ MPa
$\sigma_y = 20$ MPa
$\tau_{xy} = 30$ MPa

centre
$C = 50$ MPa

radius
$R = 42.4$ MPa

$\sigma_1 = 92.4$ MPa
$\sigma_2 = 7.6$ MPa

REMEMBER / Rotate the axes and the shear can be removed; the principal stresses cannot.